

OWEN THARP

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EDUCATION

Webb School of Knoxville, Knoxville, TN | Expected Graduation: May 2028

GPA: 4.50 weighted / 4.00 unweighted

Relevant Coursework: AP Computer Science A, 100% average and peer tutoring; Honors Quantum Computing; Honors Drone Technology and Aviation, autonomous delivery research project

TECHNICAL SKILLS

Programming: Java, Python, C++, TypeScript, React

Robotics & Embedded Systems: WPILib Java, RoboRIO, Raspberry Pi, Orange Pi, ESP32, sensor integration, control systems

CAD/CAM: Onshape, Fusion 360 CAM, SolidWorks fundamentals, CNC machining workflow

AI/ML: PyTorch, TensorFlow/Keras, CNNs, Vision Transformers, model evaluation, Google Colab, speech-to-text and text-to-speech pipelines

EXPERIENCE

Competitive Robotics, FLL, FTC, and FRC | Fall 2022 - Present

Head Programmer, FRC Team 1466

- Led development of the robot autonomous stack in WPILib Java, integrating Choreo trajectories, PIDF-controlled mechanisms, AprilTag-based 3D localization, and command-based sequencing for match-ready autonomous routines.
- Maintained localization accuracy within approximately 0.05 meters and 0.1 radians during robot operation through vision integration, pose estimation, and drivetrain feedback.
- Designed a finite-state coordination layer for scoring, intake, and mechanism handoff workflows, allowing autonomous and teleoperated routines to share the same robot-state architecture.
- Led and taught a 6-student programming group, creating and presenting 8 technical slide decks with interactive follow-up lessons on robot software, controls, localization, and debugging.
- Designed robot mechanisms in Onshape, generated CAM in Fusion 360, and manufactured CNC parts for off-season projects.
- Contributed to an FRC program recognized with the 2025 FIRST Engineering Inspiration Award and 2026 Judges Award; qualified for the FRC World Championship, competing with teams from 35+ countries.

Co-Captain, FTC Team 9934

- Co-led a 15+ member team that won the 2024 Tennessee FTC State Championship.
- Wrote autonomous and teleoperated robot control code and coordinated the workflow between mechanism design, implementation, testing, and match strategy.
- Helped translate design goals into buildable robot systems by connecting CAD, programming, and driver feedback.

PROJECTS

Harvard AI Bootcamp | Summer 2025

- Built breast-cancer image classifiers in PyTorch using a curated benign vs. malignant subset from the Kaggle Multi Cancer Dataset.
- Compared a custom CNN, ResNet18 transfer-learning model, and Vision Transformer while tuning learning rate, batch size, epoch count, image augmentation, and architecture choices.
- Reached up to 99.25% validation accuracy and visualized training and validation performance to compare convergence behavior and diagnose overfitting.

Cisco Hack Camp | Summer 2025

- Built a solo ESP32-based bidirectional Morse code translator in C++.

- Implemented visual, keyboard, and tap input modes with real-time translation and WebSocket-based networked communication.
- Selected by program mentors and Cisco employees as the top project presenter.
- Invited back as the only student mentor in the program's history.

Artificial Intelligence Visual Assistant | Independent Project, Spring 2024

- Designed and built a local Python-based AI assistant with persistent user memory stored in structured JSON.
- Integrated OpenAI Whisper, GPT-3.5 Turbo, OpenAI text-to-speech, and NVIDIA Audio2Face into a real-time conversational avatar pipeline.
- Deployed the system with multi-user support, tested it across approximately 20 users, and presented it to the K-12 school community.

LEADERSHIP, COMPETITIONS, AND AWARDS

Science Olympiad | Fall 2023 - Present

- Awarded 1st place in Division C Competitive CAD.
- Awarded 3rd place in Division C Windmill.

TMTA Mathematics Competition | 2022 - Present

- Awarded 3rd place in Precalculus and Algebra II.

STEM Mentorship and Outreach | Summer 2024 - Present

- Mentored FLL and FTC robotics teams in programming fundamentals, systems thinking, and engineering iteration.
- Served as an Engineering Mentor at Camp Webb in 2024 and 2025, instructing Sumobots and Code Build Create programs for approximately 40 lower school students.

TRAINING

America's Cutting Edge CNC Machining Training Program

- Completed ACE CNC machining training covering foundational manufacturing concepts and machining workflow.